

# MATH 70096 – Exploratory Data Analysis & Visualisation

## Assessed Coursework 1

HAND OUT: 9:00, Thursday 5 February 2026

HAND IN: 9:00, Thursday 26 February 2026

## Overview

In this coursework you will demonstrate your ability to carry out **exploratory data analysis (EDA)**, communicate findings effectively, and reason critically about data analysis workflows and diagnostic tools.

The coursework consists of **three questions**, each with different deliverables. Together, these assess both practical EDA skills and conceptual understanding of data analysis choices.

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This assignment must be attempted individually; your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework, and must abide by [Imperial College's rules] regarding academic integrity and plagiarism. The submission of output from [generative AI tools](e.g., ChatGPT) for this coursework is prohibited. Violations will be treated as an examination offence. Enabling other candidates to plagiarise your work constitutes an examination offence. To ensure quality assurance is maintained, departments may choose to invite a random selection of students to an 'authenticity interview' on their submitted assessments.

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## Submission Components

You are required to submit the following:

1. **R Markdown file (Question 1)** Implementing the exploratory analysis for Question 1. This document should be tidy, readable, and informative, and should support your conclusions rather than act as a raw record of commands.
2. **PDF report (Questions 1 and 2)** Presenting the main findings from Question 1 and the essay response for Question 2. The report must not exceed **7 pages**, excluding references and appendices.
3. **Video presentation (Question 1)** A short presentation (maximum **2 minutes**) highlighting your main findings from Question 1.
4. **R Markdown file (Question 3)** Implementing and commenting on the simulation-based diagnostic exercise for Question 3.

Further submission and file-naming instructions will be provided on Blackboard.

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## Mark Allocation Summary

| Question     | Component(s)                            | Marks       |
|--------------|-----------------------------------------|-------------|
| Question 1   | Applied EDA (R Markdown, report, video) | 28 (15,8,5) |
| Question 2   | Tidyverse essay (report only)           | 8           |
| Question 3   | Diagnostic reasoning (R Markdown)       | 14          |
| <b>Total</b> |                                         | <b>50</b>   |

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## Question 1: Applied Exploratory Data Analysis

In this question you will carry out an exploratory data analysis of a real-world dataset, focusing on identifying structure, relationships, and unusual behaviour in the data.

You are provided with a dataset containing daily records from a public bike-sharing system. Each observation corresponds to a single day, and the data include measures of bike rental demand together with weather conditions and calendar-related variables.

You should assume that your analysis will be read by a technically competent data scientist.

### Data

The dataset is based on the *Bike Sharing Dataset* and will be provided on Blackboard as part of the coursework materials. A description of the variables included in the dataset is given in Table 1.

| Variable   | Description                             |
|------------|-----------------------------------------|
| dteday     | Date of observation                     |
| cnt        | Total number of bike rentals            |
| season     | Season indicator                        |
| yr         | Year indicator                          |
| mnth       | Month                                   |
| holiday    | Whether the day is a public holiday     |
| workingday | Whether the day is a working day        |
| weathersit | Categorical weather situation indicator |
| temp       | Normalised temperature                  |
| atemp      | Normalised apparent temperature         |
| hum        | Normalised humidity                     |
| windspeed  | Normalised wind speed                   |

Table 1: Description of variables included in the bike sharing dataset.

### Task

You are required to perform an exploratory data analysis that addresses the following:

- identification and discussion of key **dependencies or relationships** in the data;
- identification and discussion of **outliers** or extreme observations;
- identification of any **anomalies or irregular structure** that merit further investigation.

Your analysis should be supported by well-designed data visualisations and clear interpretation. You are free to choose which variables to focus on and which exploratory techniques to use, but your choices should be justified.

### Deliverables

- **R Markdown file (Q1):** This should implement your exploratory analysis. The document should be tidy, readable, and informative, and should not simply be a chronological

record of commands. Commentary should explain the purpose of analyses and visualisations.

- **PDF report (Q1 section):** This should present a concise summary of your main findings from the exploratory analysis. The report should not be a knitted version of the R Markdown file; instead, it should communicate key insights and conclusions clearly. **This should be the SAME materials as presented in the video presentation.**
- **Video presentation (Q1):** A short presentation (maximum 2 minutes) summarising your main findings. The presentation should use a small number of well-chosen visualisations and focus on interpretation rather than technical detail.

## Question 2: Essay – The tidyverse approach

Write a short essay (maximum **500 words**) discussing the **strengths and weaknesses of the tidyverse approach** to data analysis.

Your discussion must refer to concrete examples, drawn either from:

- your work in Question 1, or
- other clearly described personal experience with data analysis.

Purely abstract or generic discussion is discouraged.

### Deliverable

- **Report (Q2 section only)**

## Question 3: Diagnostic Reasoning via Simulation

In this question you will explore issues of data representation and diagnostic reasoning using simulated data. You will work with a simple logistic regression model, treated as a black-box transformation from inputs to predicted class membership probabilities.

In many applied settings, continuous variables are not always used in their original form. For example, values may be discretised into bands or deliberately perturbed by adding noise, for reasons related to data quality, privacy, or standardisation. The aim of this question is to explore how such representation choices affect downstream outputs, and how these effects can be investigated using exploratory data analysis.

The focus of this question is on exploratory comparison and interpretation, not on model fitting or predictive performance.

## Simulation Setup

You will work with a simulated dataset consisting of:

- a binary response variable  $y$ ;
- a single continuous predictor  $x$ ;
- a fixed sample size of  $n = 1000$  observations.

You are provided with R code to:

- simulate the dataset;
- fit a logistic regression model of  $y$  on  $x$ ;
- extract predicted class membership probabilities for each observation.

You must use the provided code as the starting point for this question. This code is at the end of the document.

## Task

Starting from the original predictor variable  $x$ , you are required to construct **two alternative representations**:

### 1. Fixed-width discretisation

Replace each observed value of  $x$  by the **midpoint of the fixed-width interval** in which it falls. (For example, this could be achieved by partitioning the range of  $x$  into equal-width bins and mapping each observation to its bin midpoint.)

### 2. Jittered representation

Replace each observed value of  $x$  by a jittered version obtained by adding a small amount of random noise.

You should then:

- fit a logistic regression model using each representation of the predictor;
- extract the corresponding predicted class membership probabilities;
- use exploratory data analysis techniques to **compare the resulting distributions** of predicted probabilities.

You are free to choose how to carry out these comparisons. For example, you might consider:

- histograms;
- kernel density estimates;
- QQ plots (comparisons between distributions).

## Interpretation and Discussion

Your discussion should address the following points:

- how similar or different the distributions of predicted probabilities are across the three representations;
- what this suggests about the sensitivity of downstream outputs to representation choices;
- the **challenges and limitations** of the visual comparison methods you have chosen (for example, dependence on tuning choices, interpretability, or the difficulty of comparing more than two distributions);
- what can and cannot reasonably be concluded from these exploratory diagnostics.

## Implementation Hints

The following hints are provided to help you get started. They are indicative only; you are not required to use any specific implementation.

- Fixed-width discretisation may be implemented by first assigning observations to bins (for example, using `cut()`) and then replacing each value by the midpoint of its bin.
- A jittered representation may be obtained by adding small random noise to the original predictor (for example, using `jitter()` or by adding draws from a zero-mean normal distribution).
- Logistic regression models can be refitted using alternative predictors by modifying the model formula accordingly.
- Predicted class membership probabilities can be extracted using `predict()` with an appropriate response type.

You should clearly state and justify any choices you make (for example, bin widths or jitter magnitude), and reflect on how these choices may affect your conclusions.

## Deliverable

- **R Markdown file (Q3)**

This document should include the provided simulation and modelling code, your extensions, and explanatory commentary. The emphasis should be on reasoning and interpretation rather than on technical optimisation.

## Guides on Marking Criteria

Your submission will be assessed according to the following criteria:

- **Exploratory analysis and insight:** Appropriateness of analyses and visualisations; ability to identify and interpret dependencies, outliers, and anomalies.
- **Data visualisation and communication:** Clarity, accessibility, and restraint in visual design; effectiveness of communication in the report and video.
- **Workflow, code quality, and reproducibility:** Clear, tidy, and well-structured R Markdown documents; code and commentary that support understanding.
- **Conceptual understanding and critical reflection:** Balanced discussion supported by specific examples and critical judgement.
- **Diagnostic reasoning and interpretation:** Appropriate choice and justification of exploratory comparisons; correct interpretation of what diagnostics can and cannot support.
- **Presentation and adherence to instructions:** Clarity and conciseness of writing, adherence to page and time limits, and overall professionalism.

## Final Notes

There is no single correct analysis. Reasonable choices that are clearly justified will be rewarded. Unsupported claims, uncritical plotting, or excessive output without interpretation will be penalised.

You are encouraged to demonstrate judgement, restraint, and clarity throughout.

```

1 #####
2 # EDAV Coursework 1: Question 3 starter code (R)
3 #####
4
5 set.seed(123)    # set to the last three numbers of your College ID
6 n <- 1000
7
8 x <- rnorm(n, mean = 0, sd = 1)
9
10 eta <- -0.5 + 1.2 * x
11 p <- plogis(eta)
12
13 y <- rbinom(n, size = 1, prob = p)
14
15 df <- data.frame(y = y, x = x)
16
17 fit <- glm(y ~ x, data = df, family = binomial(link = "logit"))
18
19 phat <- predict(fit, newdata = df, type = "response")

```

Listing 1: R code for Question 3: simulation and model fitting